D Data
S Structures
A Algorithms

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Data Structures & Algorithms

Today's Topic: Graph Traversal

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https://sithuaung-ink.github.io/data-analyst-portfolio/

5/7/2026

Graph Traversal

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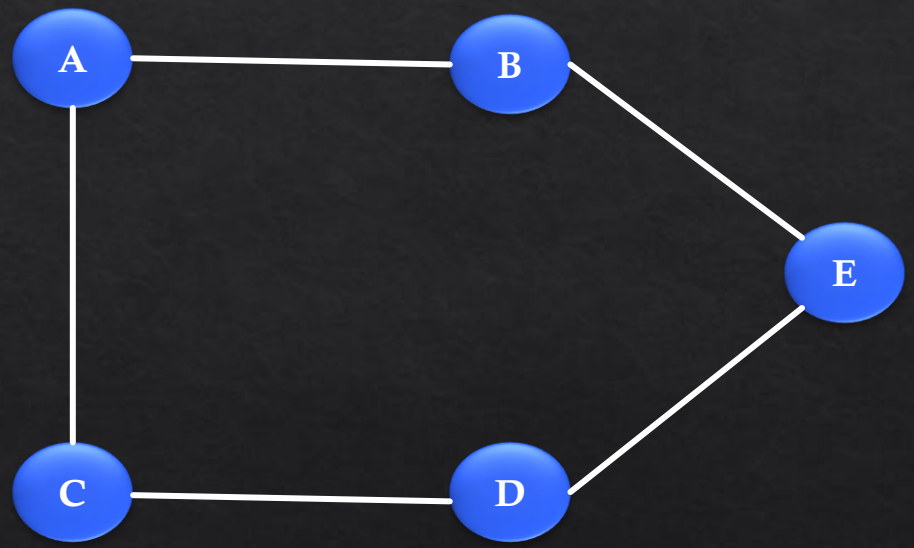
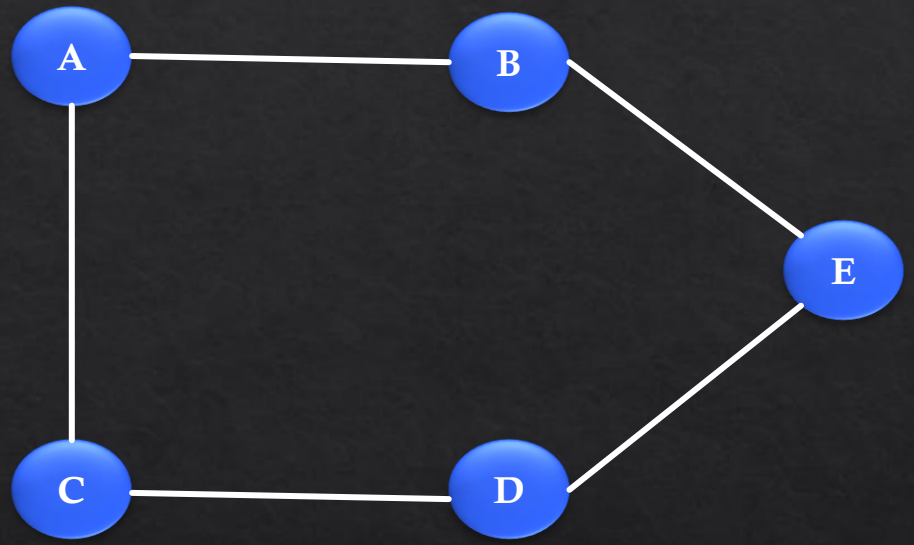
- Breadth First Search – BFS
- Depth First Search – DFS



Breadth First Search – BFS




```
{
    A: [B, C],
    B: [A, E],
    C: [A, D],
    D: [C, E],
    E: [B, D]
}
```

Breadth First Search – BFS

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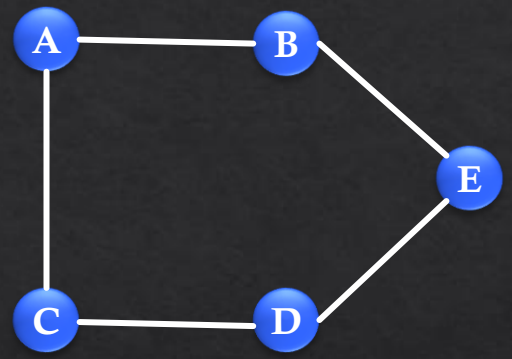
```
def bfs(self, vertex):
    visited = set()
    visited.add(vertex)
    queue = [vertex]
    while queue:
        current_vertex = queue.pop()
        print(current_vertex)
        for adjacent_vertex in self.adjacency_list[current_vertex]:
            if adjacent_vertex not in visited:
                visited.add(adjacent_vertex)
                queue.append(adjacent_vertex)
```

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{
    A: [B, C],
    B: [A, E],
    C: [A, D],
    D: [C, E],
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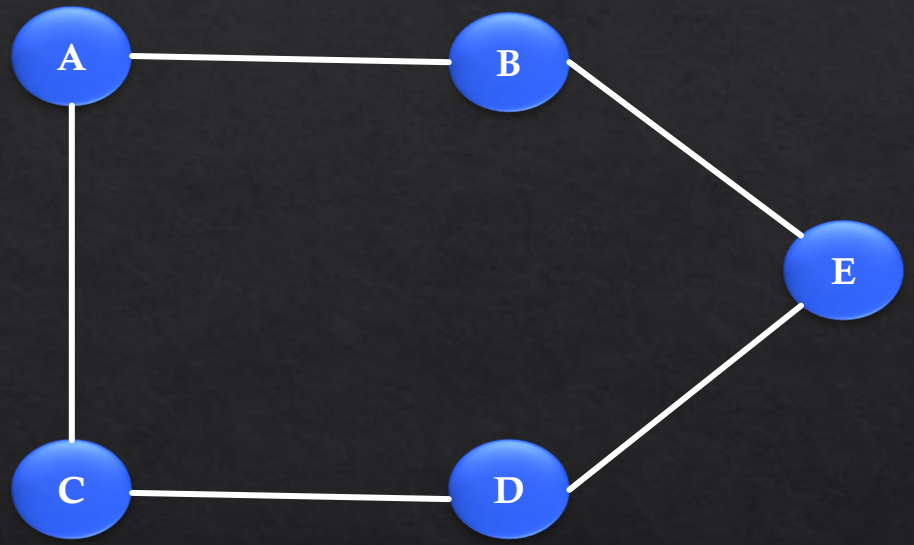
}
```

Depth First Search – DFS




```
{
    A: [B, C],
    B: [A, E],
    C: [A, D],
    D: [C, E],
    E: [B, D]
}
```

Depth First Search – DFS

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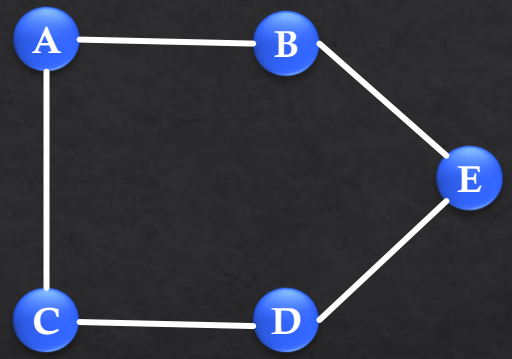
```
def dfs(self, vertex):
    visited = set()
    stack = [vertex]
    while stack:
        current_vertex = stack.pop()
        if current_vertex not in visited:
            print(current_vertex)
            visited.add(current_vertex)
        for adjacent_vertex in self.adjacency_list[current_vertex]:
            if adjacent_vertex not in visited:
                stack.append(adjacent_vertex)
```

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    C: [A, D],
    D: [C, E],
    E: [B, D]
}
```

BFS vs DFS

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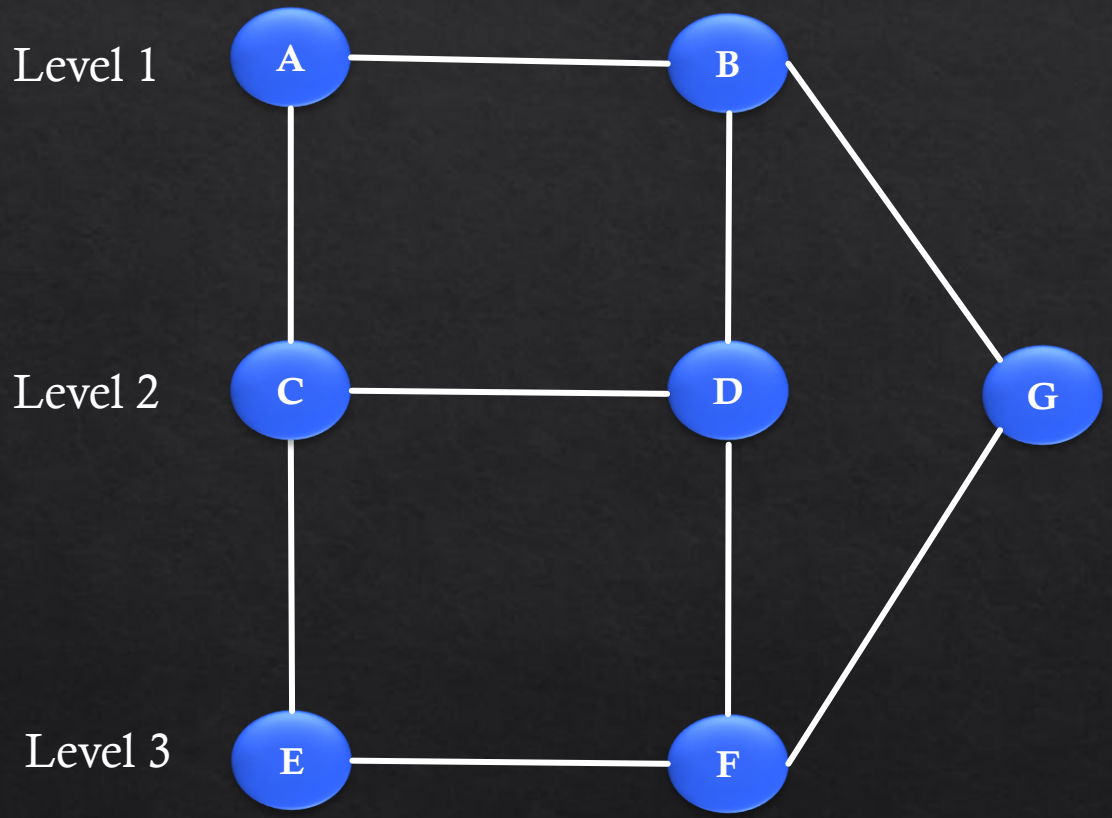
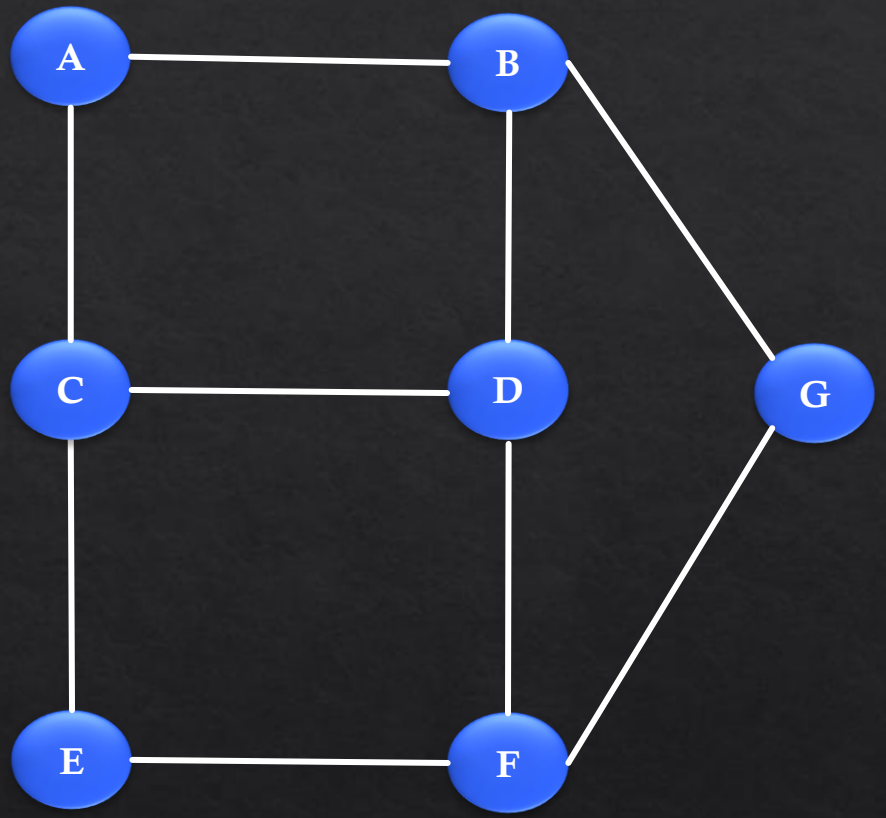


BFS



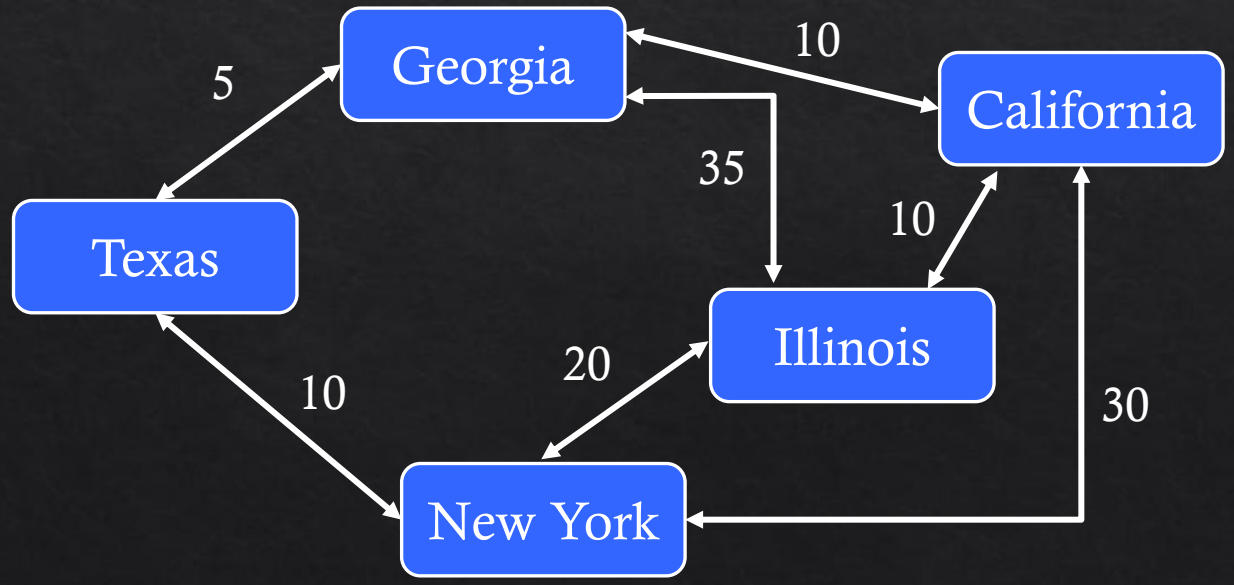
DFS

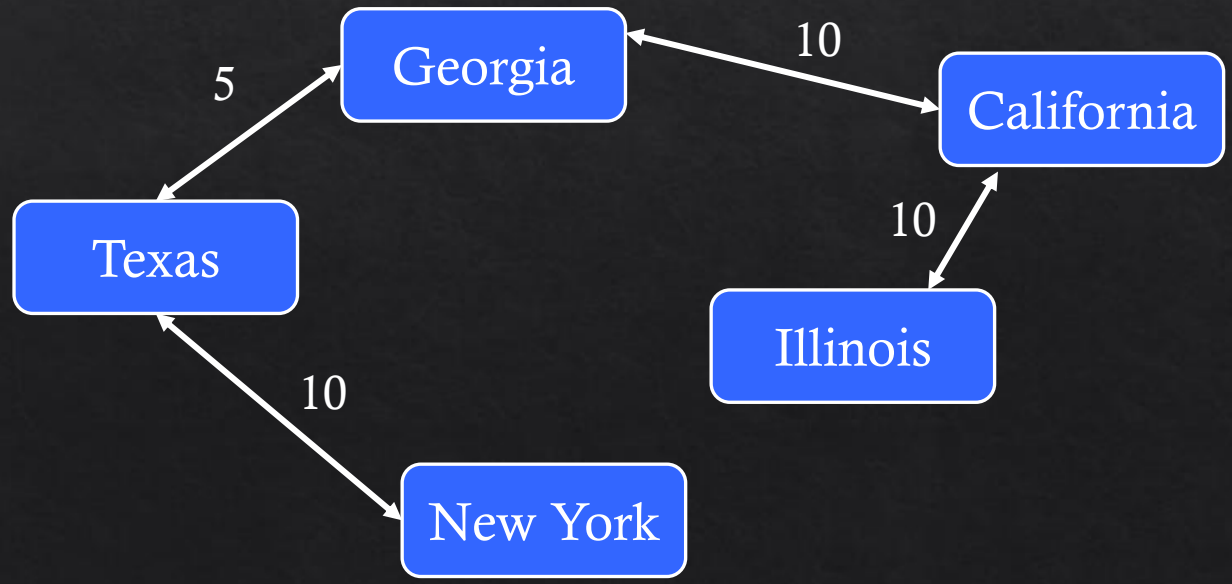
Single Source Shortest Path Problem

A single source shortest path problem is about finding a path between a given vertex and all other vertices in a graph such that the total distance between them is minimum.

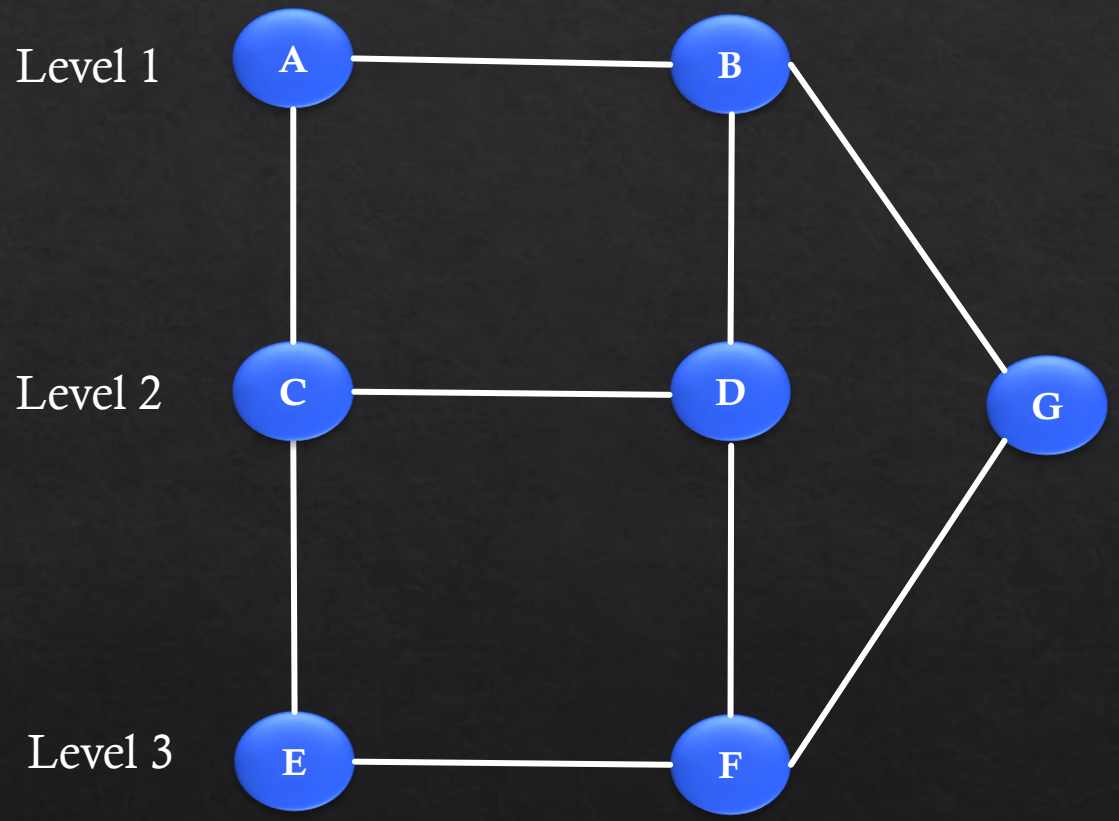
The problem:

- Five offices in different states
- Travel costs between these states are known
- Find the cheapest way from the head office to the branches in different states.





BFS



BFS

BFS – SSSPP

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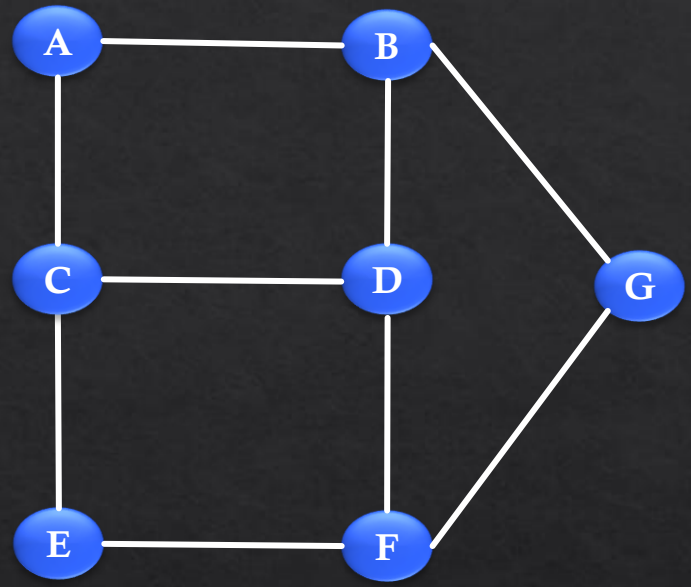
```
def bfs(self, start, end):
    queue = []
    queue.append([start])
    while queue:
        path = queue.pop(0)
        node = path[-1]
        if node == end:
            return path
        for adjacent in self.gdict.get(node, []):
            new_path = list(path)
            new_path.append(adjacent)
            queue.append(new_path)
```

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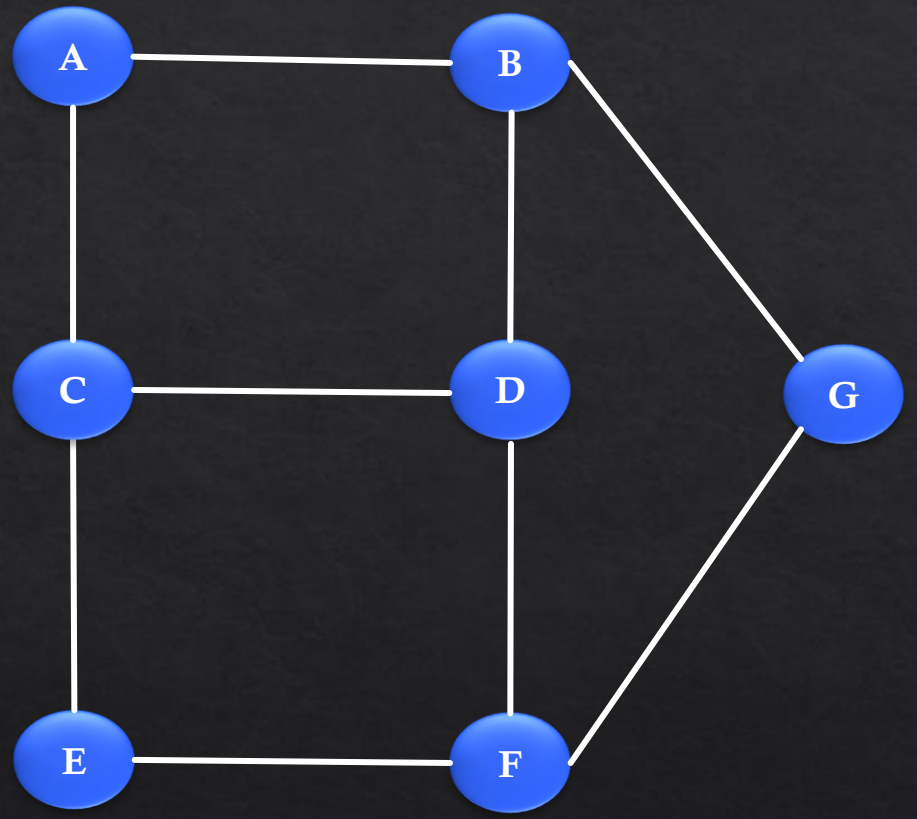
DFS – SSSPP



DFS